

SonarQube

SonarQube works like a sanitiser and helps clear bugs, vulnerabilities and code smell in the code. It supports more than 20 programming languages for static code analysis (SCA).

It is much easier to integrate static code analysis using SonarQube in Pipeline as Code. This tool helps to transform the culture of an organisation from manual to automated code review using quality gates and profiles.

Initial release	2006-07
Stable release	8.9
Written in	Java
Licence	Lesser GNU General Public License
Website	https://www.sonarqube.org/
GitHub repository	https://github.com/SonarSource/sonarqube
Features	- Support for the programming languages Java (including Android), C#, C, C++, JavaScript, TypeScript, Python, Go, Swift, COBOL, Apex, PHP, Kotlin, Ruby, Scala, HTML, CSS, ABAP, Flex, Objective-C, PL/I, PL/SQL, RPG, T-SQL, VB.NET, VB6, and XML - Quality gates - Quality profiles
How is it useful in implementing DevOps practices?	Static code analysis
Can we integrate it with Pipeline as Code?	Yes
Is a commercial flavour available?	Yes

Grafana

This open source tool is fully managed by Grafana cloud, but you can run the enterprise version on your infrastructure with self-managed options.

Grafana is one of the most popular tools when we are using Kubernetes. It is easy to install, configure and manage Grafana and monitoring metrics keeping Prometheus as a data source.

Initial release	2014
Stable release	8.3.3
Written in	Go, TypeScript
Licence	GNU Affero General Public License, version 3.0
Website	https://grafana.com/
GitHub repository	https://github.com/grafana/grafana
Features	- Panels - heatmaps to histograms - Plugins - https://grafana.com/grafana/plugins/ - Alerts - Annotations
How is it useful in implementing DevOps practices?	Visualisation of monitoring metrics
Is a commercial flavour available?	https://grafana.com/pricing/



Figure 8: Grafana dashboard

Ansible

Ansible is an open source IT automation/configuration management tool that provides support for cross-platforms. It is easy to use, and provides security and reliability for safe adoption in an organisation.

It is also popular for deployments and rolling updates in release management activities. Ansible is different from other popular configuration management tools as it is agentless. It can be integrated with Kerberos, LDAP, and other authentication management systems.

Initial release	2012
Stable release	5.0.1
Written in	Python, PowerShell, Shell, Ruby
Licence	GNU General Public License
Website	https://www.ansible.com/
GitHub repository	https://github.com/ansible/ansible
Features	- Distributed architecture - Easy integration with Docker - Supports hybrids, on-premise, or cloud environments
How is it useful in implementing DevOps practices?	- Configuration management tool - Infrastructure as Code
Can we integrate it with Pipeline as Code?	Yes
Is a commercial flavour available?	Yes https://www.ansible.com/products/pricing

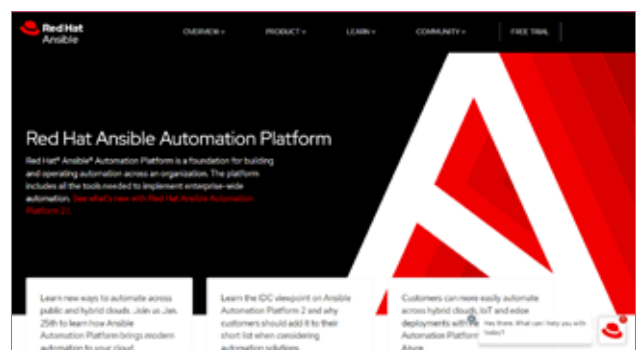


Figure 9: Ansible